

Add Employee — Gold Standard

The **single-requirement gold standard** for Phase 1. This is the bar: a senior QA architect cannot easily think of an important scenario that LevelUp missed.

Not a leaderboard. Not a score to optimise. It is the exhaustive list of scenarios an experienced QA architect would expect a test suite for **Add Employee** to contain, so we can generate, compare, and fix whatever is missing — one capability at a time.

The requirement under test

Add Employee (HR / Employee Management)

An authorized admin adds a new employee by entering **first name**, **last name**, an **employee ID**, and an **optional profile photo**. The employee ID may be **auto-generated or entered manually** and **must be unique**. On save the record is created, a **success notification** is shown, the admin is **redirected to the employee list**, and the new employee becomes **searchable by ID and by name**.

How to read this file

- Every scenario a senior QA would expect is listed under its category.
 - **Priority** — `Critical` (a defect here is a serious escape) → `High` → `Medium` → `Low` (polish / nice-to-have).
 - **Detection keywords** — how the audit harness (`scripts/add-employee-audit.ts`) decides the generated suite covers this scenario. Any one phrase present in a generated scenario's title/objective/risk-area (token-stemmed, via the shared `coverage-match` matcher) counts as covered. These describe the concept — they are **not** a script to match verbatim.
 - **Grounding** — `explicit` = the requirement states it directly; `expected` = a senior QA expects it for an HRM Add-Employee form even though the one-paragraph requirement does not spell it out. A miss on an `expected` scenario is a real conversation ("should the generator raise this?"), not automatically a defect.
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1. Functional — core creation paths

ID	Scenario	Priority	Grounding	Detection keywords
FUN-01	Create employee with all fields valid (incl. photo)	Critical	explicit	create employee, valid employee, add employee, create a record with valid
FUN-02	Create with mandatory fields only (photo omitted — it is optional)	High	explicit	minimum required, mandatory fields only, required fields only, without a photo, optional photo omitted
FUN-03	Create with an auto-generated employee ID	High	explicit	auto-generated id, auto generated id, auto-generated id
FUN-04	Create with a manually-entered employee ID	High	explicit	manually entered id, manual id, enter id manually
FUN-05	Created employee is retrievable / appears in the employee list	High	explicit	appears in the employee list, retrievable, record is retrievable, view an existing record

2. Validation — field-level, one scenario per field

ID	Scenario	Priority	Grounding	Detection keywords
VAL-01	First name blank is rejected	Critical	explicit	first name blank , first name required , missing first name , first name empty
VAL-02	Last name blank is rejected	Critical	explicit	last name blank , last name required , missing last name , last name empty
VAL-03	Both names blank is rejected	High	explicit	both names blank , both fields blank , all required fields blank
VAL-04	Employee ID blank (manual entry) is rejected	High	explicit	employee id blank , id required , missing employee id , id empty
VAL-05	First name whitespace-only is rejected (trimmed to blank)	High	expected	first name whitespace , whitespace-only first name
VAL-06	Last name whitespace-only is rejected (trimmed to blank)	High	expected	last name whitespace , whitespace-only last name
VAL-07	Invalid employee ID format is rejected	High	explicit	invalid id format , malformed id , invalid employee id format

ID	Scenario	Priority	Grounding	Detection keywords
VAL-08	Numeric / digit characters in a name are handled per rule	Medium	expected	numbers in name , numeric name , digits in first name
VAL-09	Field-level validation errors are shown inline	Medium	expected	inline validation error , field-level error , validation message shown

3. Business Rules — the logic that guards the data

ID	Scenario	Priority	Grounding	Detection keywords
BIZ-01	Duplicate employee ID is rejected (uniqueness)	Critical	explicit	duplicate , already exists , unique constraint , existing id
BIZ-02	Auto-generated ID is unique	High	explicit	auto-generated id is unique , generated id unique , unique auto
BIZ-03	Manually-entered ID accepted when unique	Medium	explicit	manual id accepted , unique manual id
BIZ-04	Employee ID uniqueness enforced case-insensitively	Low	expected	case-insensitive id , case insensitive unique
BIZ-05	Re-submitting the same form does not create a duplicate record	High	expected	double submit , duplicate submission , resubmit , idempotent create

4. Boundary — at, below and above every limit

ID	Scenario	Priority	Grounding	Detection keywords
BND-01	First name at maximum length is accepted	High	expected	first name max length , max length name accepted , name at maximum length
BND-02	First name one character over maximum is rejected	High	expected	name exceeds max length , name too long , over maximum length
BND-03	Employee ID at maximum length is accepted	Medium	expected	id max length , employee id maximum length
BND-04	Employee ID exceeding maximum length is rejected	High	expected	id too long , id exceeds max length , employee id over maximum
BND-05	Single-character name is accepted (minimum boundary)	Medium	expected	single character name , minimum length name , one character name
BND-06	Leading zeros in employee ID are preserved	Medium	expected	leading zero
BND-07	Unicode / accented characters in names are accepted (José, O'Brien)	Medium	expected	unicode name , accented name , special characters in name , non-ascii name
BND-08	Emoji in a name field is handled per rule	Low	expected	emoji

5. Security — hostile input must never win

ID	Scenario	Priority	Grounding	Detection keywords
SEC-01	SQL-injection string in a name field is neutralised	Critical	expected	sql injection , injection
SEC-02	XSS / script payload in a name field is escaped (not executed)	Critical	expected	xss , script injection , cross-site scripting , script payload
SEC-03	SQL-injection string in the employee ID is neutralised	High	expected	injection in id , sql injection id
SEC-04	HTML injection in a field is sanitised on display	Medium	expected	html injection , markup injection
SEC-05	An executable disguised as an image is rejected on upload	High	expected	spoofed , disguised , executable disguised , content type mismatch

6. Authorization — only the right user, only the right way

ID	Scenario	Priority	Grounding	Detection keywords
AUZ-01	Authorized admin can create an employee	High	explicit	authorized , admin can create , permitted user
AUZ-02	Unauthorized / non-admin user is blocked	Critical	explicit	unauthorized , forbidden , access denied , no permission , not allowed
AUZ-03	Direct URL / API access to the create endpoint is authorization-checked	High	expected	direct url , direct api , endpoint access control , bypass the ui
AUZ-04	Unauthenticated user is redirected to login	Medium	expected	unauthenticated , redirected to login , not logged in

7. Upload — the optional profile photo

ID	Scenario	Priority	Grounding	Detection keywords
UPL-01	Valid image (jpg/png) uploads and is associated with the record	Medium	explicit	valid file upload , valid photo , valid image , file upload is accepted
UPL-02	Save succeeds with no photo (photo is optional)	High	explicit	without a photo , photo is optional , no photo , optional photo omitted
UPL-03	Unsupported file type is rejected	High	expected	invalid file type , unsupported format , wrong format , disallowed type
UPL-04	Oversized image is rejected at the size boundary	High	expected	oversized , large file , file too large , exceeds max size
UPL-05	Image at exactly the maximum size is accepted	Medium	expected	at the limit , at maximum size , at the size boundary
UPL-06	Corrupted / truncated image is handled gracefully	Medium	expected	corrupted , corrupt image , truncated file , malformed file
UPL-07	Replace / re-upload photo before saving	Low	expected	replace photo , re-upload , change photo

8. Navigation — where the user lands

ID	Scenario	Priority	Grounding	Detection keywords
NAV-01	Success notification / confirmation is shown on save	High	explicit	success notification, success message, confirmation shown
NAV-02	Admin is redirected to the employee list after save	Medium	explicit	redirect, navigates to, employee list
NAV-03	Cancel discards input and returns to the list without saving	Medium	expected	cancel discards, cancel returns, cancel without saving
NAV-04	Navigating away with unsaved changes warns the user	Low	expected	unsaved changes, navigate away warning

9. Search — post-create discoverability

ID	Scenario	Priority	Grounding	Detection keywords
SCH-01	New employee is immediately searchable by employee ID	High	explicit	search by id , searchable by id
SCH-02	New employee is immediately searchable by name	High	explicit	search by name , search-able by name
SCH-03	Partial-name search returns the new employee	Medium	expected	partial name , partial search , partial match
SCH-04	Search is case-insensitive	Medium	expected	case-insensitive search , case insensitive search
SCH-05	New record is findable with no reindex delay	Medium	explicit	immediately searchable , no reindex delay , searchable straight away

10. Data Integrity — what is stored is what was entered

ID	Scenario	Priority	Grounding	Detection keywords
DAT-01	Record is persisted and retrievable after save	High	explicit	persisted , record created , stored , saved to
DAT-02	All entered values are stored exactly (no truncation / mutation)	Medium	expected	stored exactly , no truncation , values preserved , data integrity
DAT-03	Uploaded photo is persisted and retrievable	Low	expected	photo persisted , image stored , photo retrievable
DAT-04	Concurrent creation of the same ID — only one succeeds	Medium	expected	concurrent , simultaneous , race condition

11. Integration — the record does not live alone

ID	Scenario	Priority	Grounding	Detection keywords
INT-01	New employee propagates to dependent views (directory / list)	Medium	expected	propagates , dependent module , appears in directory
INT-02	Success notification is delivered per spec (toast / email)	Low	expected	notification delivered , email notification , toast
INT-03	An audit-log entry is created for the new employee	Low	expected	audit log , audit trail , audit entry

12. Usability — the human using the form

ID	Scenario	Priority	Grounding	Detection keywords
USA-01	Mandatory fields are visually marked	Low	expected	mandatory field marked , re-quired field indicator , marked as re-quired
USA-02	Validation errors are clear and actionable	Medium	expected	clear error message , actionable error , helpful validation
USA-03	Form is keyboard-navigable with associated labels (a11y)	Low	expected	keyboard navigable , accessibility , associated label , a11y

Scoreboard (filled by the audit harness)

Run `npx ts-node scripts/add-employee-audit.ts` . It generates the current Test Case Lab output for this requirement, checks each scenario above, and prints a per-category ✓/✗ table plus the ordered missing list. The number that matters is not a percentage — it is “what would a senior QA notice is missing,” worst-priority first.

Environment note (honesty). No live LLM/DB is available in this workspace, so “generated output” here is the deterministic **planner ceiling** — every scenario the Knowledge Base + planner can emit for this requirement (all coverage families, Deep on). The LLM stage can only word up these scenarios, never add ones the planner didn’t plan, so the ceiling is an honest upper bound on the shipped suite. Prompt/LLM-stage losses cannot be observed here and are not claimed as fixed.

Cycle log — measured, not estimated

Each cycle: generate → compare → for every miss ask which stage lost it → fix only that stage → re-run this same audit. Numbers are from real runs.

Cycle	Fix (stage)	Gold covered at ceiling	Critical miss	High miss
Baseline	—	31/62	2	9
1 — Security	KB: crud had zero injection obligations → added SQL + XSS (universal)	35/62	0	8
2 — Field-aware	Planner: validation/boundary were collapsed into one generic obligation each → Phase 3c expands per named field	45/62	0	2
3 — Depth	KB: double-submit, endpoint authz, unauthenticated redirect, partial + case-insensitive search, propagation, cancel	53/62	0	0

Where the losses were: every Add-Employee miss traced to the deterministic backbone — either a missing **Knowledge-Base obligation** (Cycles 1 & 3) or a **planner** that collapsed per-field checks into one generic case (Cycle 2). None required a new engine or architecture; all fixes live in the canonical crud KB category and the existing planner, so they transfer to every CRUD requirement (Create Employee 88→100%, Edit Employee 77→86%, User Profile 65→70% on the sealed cross-requirement scorer — zero regressions).

The 9 remaining gold rows are honest, documented gaps, all Low/Medium quality attributes a one-paragraph functional requirement does not imply: audit-log entry, notification channel (toast/email), keyboard/allly, mandatory-field marking, emoji handling, unsaved-changes warning, photo-persistence, same-ID concurrency, and “errors are clear and actionable”. Manufacturing keyword rules for these would be the non-scaling trap this exercise exists to avoid — they are left visible in the worklist rather than gamed.